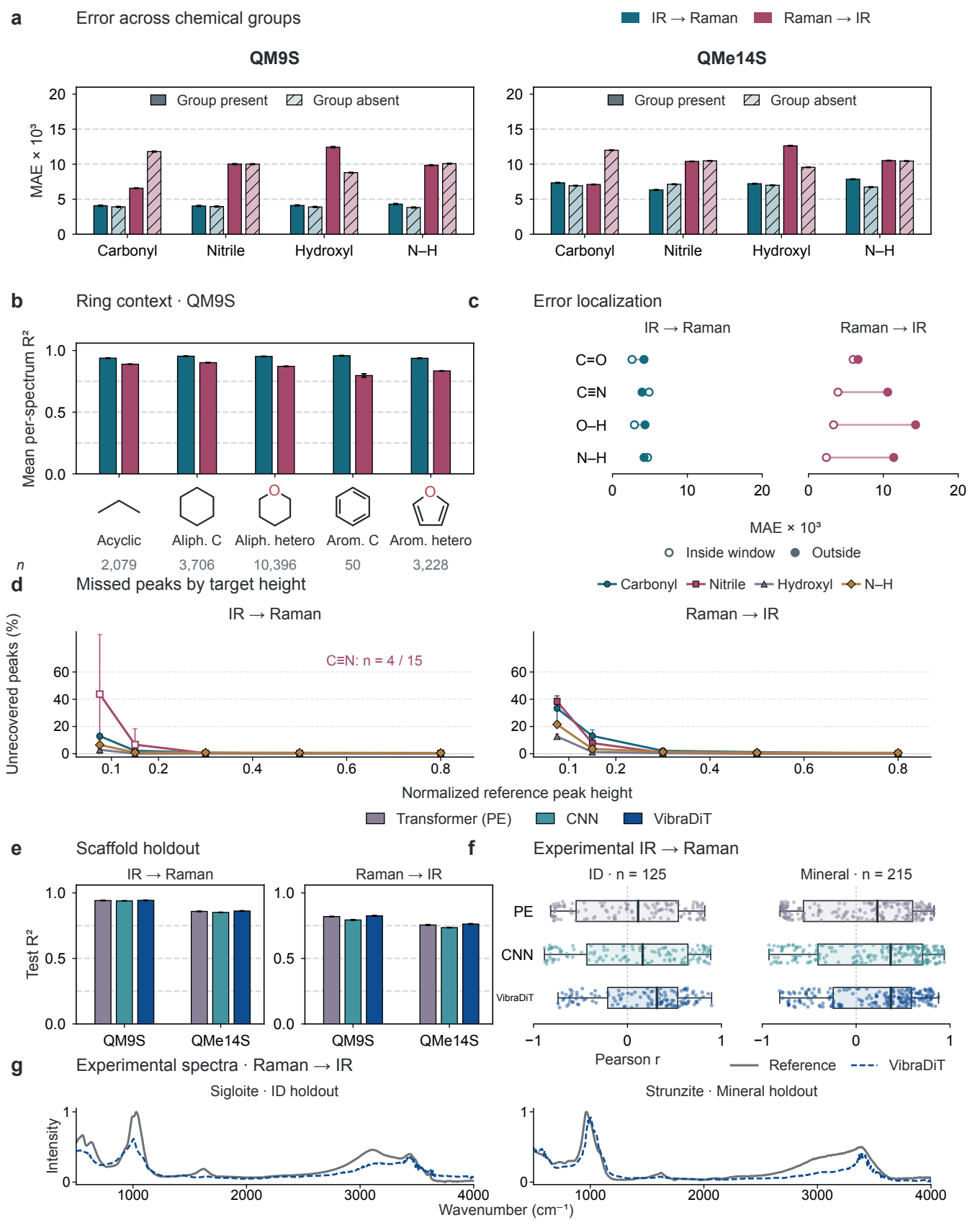

a
Error across chemical groups
IR → Raman
Raman → IR
QM9S
QMe14S
Group present
Group absent
MAE × 10³
Carbonyl
Nitrile
Hydroxyl
N–H
b
Ring context · QM9S
Mean per-spectrum R²
Acyclic
Aliph. C
Aliph. hetero
Arom. C
Arom. hetero
n
2,079
3,706
10,396
50
3,228
c
Error localization
C=O
C≡N
O–H
N–H
MAE × 10³
Inside window
Outside
d
Missed peaks by target height
Unrecovered peaks (%)
C≡N: n = 4 / 15
Normalized reference peak height
Transformer (PE)
CNN
VibraDiT
e
Scaffold holdout
Test R²
QM9S
QMe14S
f
Experimental IR → Raman
ID · n = 125
Mineral · n = 215
PE
CNN
VibraDiT
Pearson r
Reference
g
Experimental spectra · Raman → IR
Sigloite · ID holdout
Strunzite · Mineral holdout
Intensity
Wavenumber (cm⁻¹)